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3.2. Two-Stage Basis Closure: Pre-Lock and Post-Lock

The analysis of the operator basis naturally separates into two stages, distinguished by whether the dynamical "trace-lock" from the equations of motion (Result C1, proven in Sec. 4) has been imposed.

The Pre-Lock Basis

Before enforcing the equations of motion, the three surviving monomials form a complete basis for all relevant scalar densities. We denote these basis operators as:

$$\textbf{Pre-lock basis}: I_1 \equiv \Pi_{\rm PT}\big[T^A{}_{BC}T_A{}^{BC}\big], \quad I_2 \equiv \Sigma_\epsilon \equiv \Pi_{\rm PT}\big[(\partial\epsilon)^2\big], \quad I_3 \equiv \Pi_{\rm PT}\big[T_\mu \partial^\mu \epsilon\big]. \tag{5}$$

The Post-Lock Basis

After solving the Palatini equations of motion (Sec. 4), we find the algebraic trace-lock condition $T_\mu = 3\eta\,\partial_\mu\epsilon$. This condition is not a choice but a dynamical result. Applying it to the pre-lock basis causes a collapse: the mixing term I_3 is no longer independent.

$$I_3 \xrightarrow{\text{C1: } T_\mu = 3\eta\,\partial_\mu\epsilon} \Pi_{\rm PT}\big[(3\eta\,\partial_\mu\epsilon)\partial^\mu\epsilon\big] = 3\eta\,\Sigma_\epsilon = 3\eta\,I_2. \tag{6}$$

The basis of independent operators thus reduces. For convenience, we define the standard torsion invariant $I_T \equiv -\frac{1}{4}I_1$ and express the post-lock basis in terms of $\{I_T, \Sigma_\epsilon\}$.

Post-lock basis (after C1): The basis reduces to two linearly dependent operators, e.g., $\{I_T, \Sigma_\epsilon\}$. (7)

The following lemma formalizes this two-stage structure, with its proof relying on standard tensor decomposition and the self-adjointness of the projector under the A1–A5 posture.

Lemma 3 (Two-stage closure)**.** *Under assumptions A1–A5, any PT-even quadratic scalar density $\mathcal{O}_{(2)}(T, \partial\epsilon)$ with at most one derivative per factor can be expanded on the pre-lock basis $\{I_1, I_2, I_3\}$ up to a total divergence. After enforcing the C1 trace-lock, it can be expanded on the post-lock basis $\{I_T, \Sigma_\epsilon\}$.*

3.3. The Action Skeleton and Order of Operations

The closure of the operator basis allows us to write down the most general action at this order. The minimal bulk action skeleton is first written in the pre-lock basis:

$$S_{\rm bulk} = \int d^4x\,\sqrt{-g}\Big[\tfrac{M_{\rm Pl}^2}{2}\,R(e,\omega) + \alpha_1\,I_1 + \alpha_2\,I_2 + \alpha_3\,I_3\Big], \tag{8}$$

where $\alpha_{1,2,3}$ are free real coefficients. After applying the C1 lock, the effect of I_3 is absorbed into the coefficient of $I_2 \equiv \Sigma_\epsilon$, yielding the post-lock action:

$$S_{\rm bulk}^{(\rm post-lock)} = \int d^4x\,\sqrt{-g}\Big[\tfrac{M_{\rm Pl}^2}{2}\,R(e,\omega) + \alpha_1\,I_1 + \big(\alpha_2 + 3\eta\,\alpha_3\big)\,\Sigma_\epsilon\Big]. \tag{9}$$

As will be proven in Sec. 4, the C1 lock further imposes the crucial relation $I_T = -6\eta^2\,\Sigma_\epsilon$, revealing that the post-lock basis contains only one independent degree of freedom at this order.

Order of Operations and Consistency

This two-stage structure defines a clear and consistent logical pipeline for our analysis. The scalar projector $\Pi_{\rm PT}$ acts at the level of densities, and by Lemma 1, we may *project then vary*. The trace lock, in contrast, is an *algebraic* enforcement at the level of the equations of motion; it is not a projection and introduces no new canonical pairs. The correct procedure is therefore:

(i) Enumerate & project $\Rightarrow$ Pre-lock basis $\{I_1, I_2, I_3\}$ $\quad\Longrightarrow\quad$ (ii) Impose C1/trace lock $\Rightarrow$ Use post-lock basis $\{I_T, \Sigma_\epsilon\}$.

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